

TEMPERATURE MONITORING SYSTEM

Arduino UNO + LM35 + 16×2 I2C LCD

TASK 1 • PROJECT README & IMPLEMENTATION GUIDE

MICROCONTROLLER	SENSOR	DISPLAY	LOGGING
Arduino UNO	LM35	16×2 I2C LCD	Serial @ 9600

1. Project Overview

This project demonstrates a simple embedded temperature-monitoring system. An LM35 temperature sensor produces an analog voltage proportional to temperature. The Arduino UNO reads this voltage through analog pin A0, converts it into degrees Celsius, displays the live value on a 16×2 I2C LCD, and sends each reading to the Serial Monitor for logging.

2. Objective

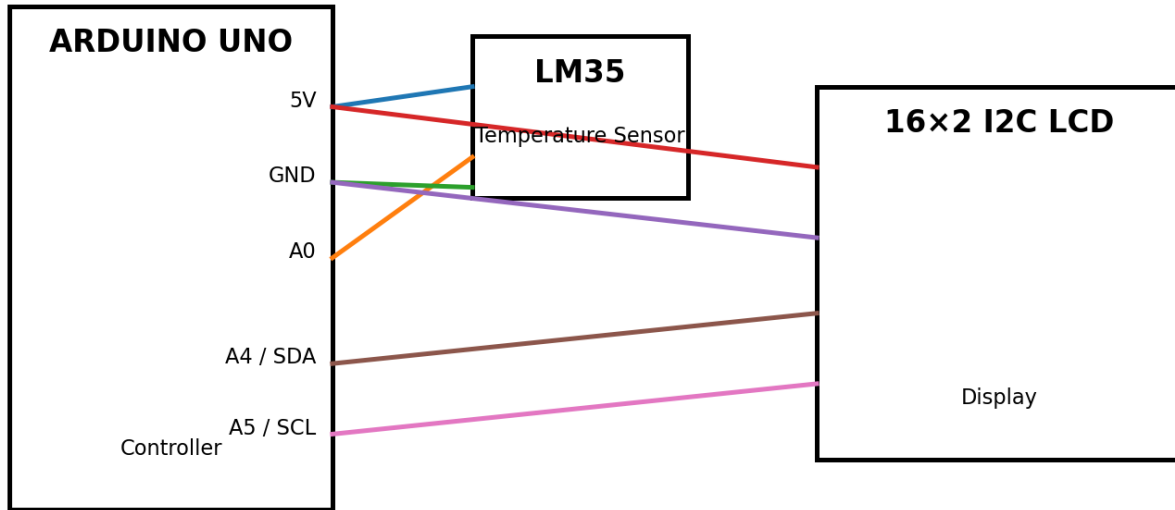
- Measure temperature using an LM35.
- Process the sensor signal with an Arduino UNO.
- Display the live temperature on a 16×2 I2C LCD.
- Transmit readings through the serial interface.
- Demonstrate a sensor → controller → display → logging workflow.

3. Components Required

Component	Qty.	Purpose
Arduino UNO	1	Main controller and ADC
LM35	1	Temperature sensing
16×2 I2C LCD	1	Live display
Breadboard	1	Circuit assembly
Jumper wires	As required	Connections
USB cable	1	Programming/ power
Arduino IDE	1	Code development
LiquidCrystal_I2C	1	LCD control

4. Circuit & Wiring Diagram

TEMPERATURE MONITORING SYSTEM



Verify LM35 pin order from your sensor datasheet.

Device	Pin	Arduino UNO
LM35	VCC	5V
LM35	OUT	A0
LM35	GND	GND
I2C LCD	VCC	5V
I2C LCD	GND	GND
I2C LCD	SDA	A4 / SDA
I2C LCD	SCL	A5 / SCL

Important: LM35 packages can have different physical pin arrangements. Confirm VCC, OUT and GND from the exact sensor datasheet before powering the circuit.

5. Working Principle

- Sensing** – The LM35 senses temperature and generates an analog output voltage.
- Reading** – The Arduino UNO converts the analog signal at A0 using its 10- bit ADC.
- Conversion** – The sketch converts the ADC value to voltage and then to °C using approximately 10 mV/°C.
- Display** – The calculated temperature is written to the LCD through I2C.
- Logging** – The same reading is printed to Serial Monitor every second at 9600 baud.

6. Arduino Program

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
const int TEMP_PIN = A0;
void setup() {
  Serial.begin(9600);
```

```

lcd.init();
lcd.backlight();
lcd.setCursor(0, 0);
lcd.print("Temperature:");
lcd.setCursor(0, 1);
lcd.print("Starting...");
delay(1000);
lcd.clear();
}

void loop() {
  int rawValue = analogRead(TEMP_PIN);
  float voltage = rawValue * (5.0 / 1023.0);
  float temperatureC = voltage * 100.0;
  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(temperatureC, 1);
  lcd.write((byte)223);
  lcd.print("C ");
  lcd.setCursor(0, 1);
  lcd.print("Raw: ");
  lcd.print(rawValue);
  lcd.print(" ");
  Serial.print("Temperature: ");
  Serial.print(temperatureC, 1);
  Serial.print(" C | ADC: ");
  Serial.println(rawValue);
  delay(1000);
}

```

7. Software Setup & Upload

1. Open Arduino IDE.
2. Connect Arduino UNO by USB.
3. Install LiquidCrystal_I2C from Library Manager if needed.
4. Select Tools → Board → Arduino UNO.
5. Select the correct USB/COM port.
6. Paste the sketch and click Verify.
7. Click Upload.
8. Open Serial Monitor and select 9600 baud.
9. Observe the LCD and Serial Monitor.

8. Expected LCD Output

Temp: 27.4°C
Raw: 560

9. Expected Serial Monitor

Temperature: 27.4 C | ADC: 560
Temperature: 27.5 C | ADC: 562
Temperature: 27.5 C | ADC: 561

10. Testing & Troubleshooting

Check that the LCD backlight turns on.

Confirm the LCD shows a temperature value.

Confirm Serial Monitor prints once per second.

If LCD is blank, check VCC/GND, SDA=A4, SCL=A5 and contrast.

If needed, check the I2C address; 0x27 is common and 0x3F is another common address.

If temperature is unrealistic, verify LM35 orientation and OUT→A0.

If Serial Monitor is empty, check the COM port and 9600 baud.

11. Real- World Project References

These reference images make the report more visually engaging. They show the same type of Arduino + LM35 + LCD arrangement used in this task. Use your own circuit photo in the evidence section as the proof of your completed build.

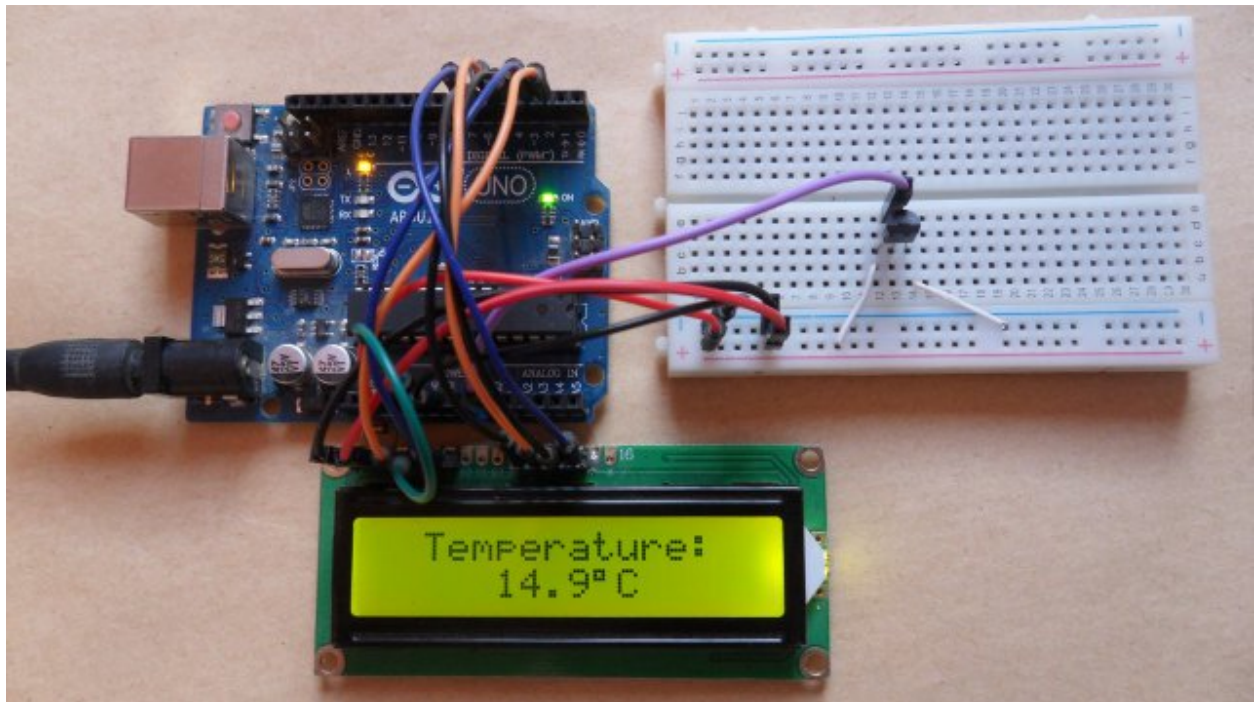


Figure 1 – Real- world Arduino/LM35 temperature display reference.

Source: Simple- Circuit, "Arduino and LM35 temperature sensor interfacing."

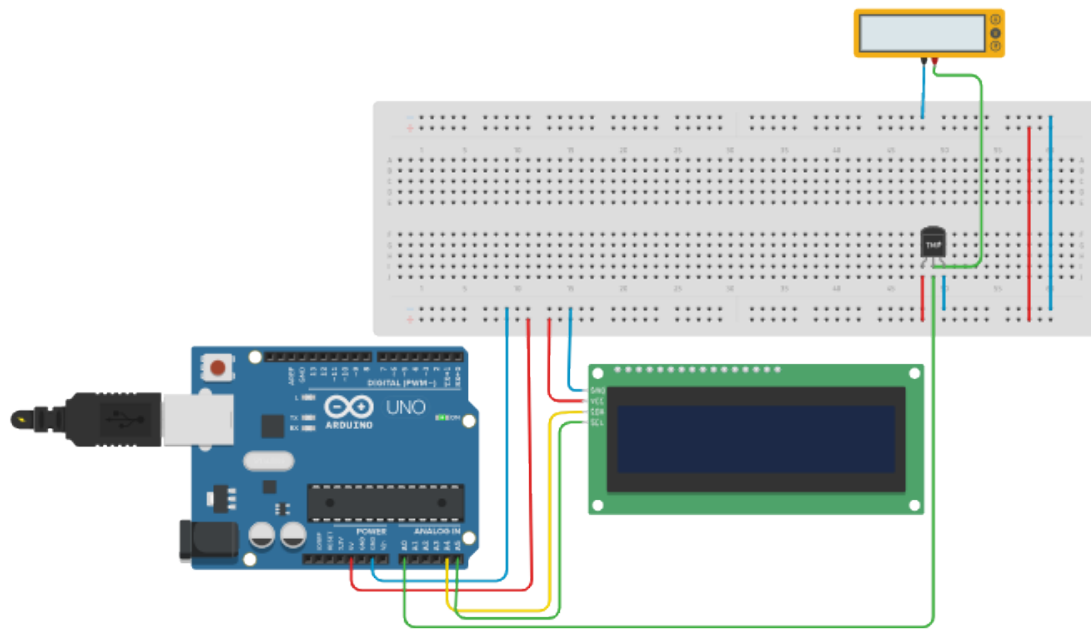


Figure 2 — Tinkercad- style circuit reference showing Arduino, breadboard and 16x2 I2C LCD.

Source: Tinkercad circuit reference, "Medir temperatura con LM35 y LCD 16x2 I2C."

DESIGN TIP

For your final submission, replace the reference images with your own clear photos wherever possible. A good evidence photo should show the Arduino UNO, LM35, breadboard, LCD and jumper wires in one frame, with the LCD powered on and displaying a live value.

12. Evidence to Attach

Circuit photograph/ screenshot.

Arduino IDE code screenshot.

Serial Monitor screenshot.

30–60 second video showing the live LCD reading.

Video link in the section below.

13. Video Submission

Video Link: _____

Recommended video flow: show the complete circuit → power the Arduino → show the LCD reading → show the Serial Monitor.

14. Result

The temperature-monitoring system reads the LM35 output with an Arduino UNO, displays the calculated temperature on a 16x2 I2C LCD, and logs the readings through the Serial Monitor.

15. Submission Checklist

Item	Status
Circuit working	Completed
Sketch uploaded	Completed
LCD live value verified	Completed
Serial logging verified	Completed
Circuit screenshot attached	Completed
IDE/ Serial screenshot attached	Completed
30–60s video recorded	Completed
Video link added	Completed